

ESP32-Based Voltage and Overcurrent Protector

Technical Design & Implementation Report

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1. INTRODUCTION

This project is a low-voltage demonstration of a voltage and overcurrent protector using a 30-pin ESP32. Two potentiometers imitate voltage and current readings. The system shows the values on a 16x2 LCD and a Wi-Fi webpage, and uses LEDs to show whether the load is ON or OFF.

2. SYSTEM ARCHITECTURE

2.1 Functional Design

The ESP32 reads the two potentiometers, compares their values with the selected limits, and updates the protection state. Three pushbuttons allow local control, while the ESP32 Wi-Fi dashboard allows monitoring, changing limits, and resetting a fault.

2.2 Finite State Machine (FSM) – 4 states

INIT: Checks the readings when the device starts.

NORMAL: Green LED ON; readings are within the set limits.

TRIPPED: A fault is detected; red LED ON and the fault stays latched.

RESET_READY: Readings are safe for 1 second; red LED blinks until reset.

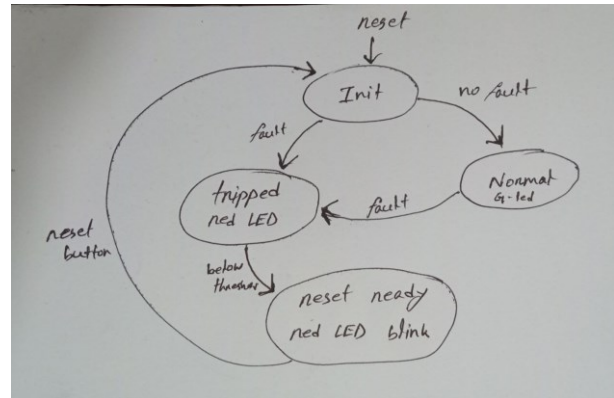


Figure 1. Hand-drawn FSM design

3. HARDWARE INTEGRATION

3.1 Pin Configuration

Voltage input: GPIO34 (ADC1)

Current input: GPIO35 (ADC1)

Green / red LEDs: GPIO25 / GPIO26

MODE / UP / DOWN: GPIO18 / GPIO19 / GPIO23

LCD (I2C address 0x27): SDA 21, SCL 22

Computer link: ESP32 Wi-Fi Access Point and dashboard

Potentiometers simulate the inputs; blue LED represent load and red one represents fault status.

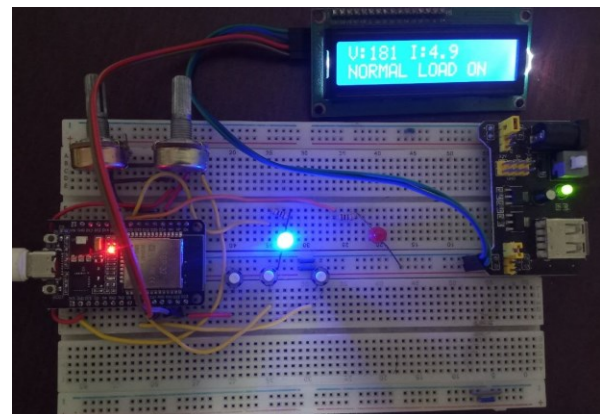


Figure 2. Assembled ESP32 hardware

4. SOFTWARE DETAILS

4.1 Critical Implementation

The ESP32 checks the inputs about every 30 ms and refreshes the LCD about every 250 ms. The starting limits are 250 V and 8 A (simulated). After a trip, values must stay below the recovery levels for 1 second before reset becomes available. Short MODE presses change screens; UP and DOWN edit limits; holding MODE resets a safe system.

4.2 Execution Order

Each loop reads buttons, checks the inputs and FSM, updates LEDs, handles Wi-Fi requests, and refreshes the LCD. The webpage and physical buttons use the same settings.

5. RESULTS

During the breadboard demonstration, turning the potentiometers changed the displayed readings, faults switched the LEDs, and the system waited for a reset after recovery. The LCD, pushbuttons, and Wi-Fi dashboard also worked for monitoring and settings.

6. CONCLUSION

The prototype demonstrates an FSM-based protection controller with local and Wi-Fi control. It uses simulated sensor readings and LEDs; it is not a real mains-voltage protection device.